Jackie Lok

CURRICULUM VITAE — LAST UPDATED NOV 2025

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ORFE Department
Princeton, NJ 08544, USA

Education

Princeton University, *Princeton, NJ, USA*
Ph.D. in Operations Research and Financial Engineering — 2021–2026 (expected)
Adviser: Elizaveta Rebrova

M.A. in Operations Research and Financial Engineering — 2023

UNSW Sydney, *Sydney, Australia*
Bachelor of Science (Honours) in Pure Mathematics — 2020
with First Class Honours and the University Medal
Supervisor: Catherine Greenhill
WAM: 97.16/100

Bachelor of Actuarial Studies (Co-op) in Mathematics with Distinction — 2016–2019
WAM: 95.65/100

Wharton School, University of Pennsylvania, *Philadelphia, PA, USA*
International Exchange Semester, GPA: 4.00/4.00 — 2017

Research Interests

My research interests broadly lie at the intersection of probability, statistics, and optimisation. I am interested in analyzing randomized algorithms in numerical linear algebra, machine learning, and data science, with a focus on developing theory and tools that allow us to better understand and work with large-scale or high-dimensional data.

Awards & Honours

- Quad Fellowship, Schmidt Futures — 2023
- Richard Stillwell '21 *24 and Agnes Newhall Stillwell Fellowship, Princeton University — 2021
- University Medal in Pure Mathematics, UNSW Sydney — 2020
- H.C. & M.E. Porter Memorial Scholarship, UNSW Sydney — 2020
- The Faculty of Science Prize for Honours Year Science, UNSW Sydney — 2020
- The George Szekeres Prize, UNSW Sydney — 2019
- The Head of School's Prize, UNSW Sydney — 2019
- UNSW Co-op Scholarship in Actuarial Studies — 2016–2019
- UNSW Scientia Scholarship — 2016–2019
- Harry Manson International Exchange Scholarship, UNSW Sydney — 2017

Travel grants/awards

SIAM Student Travel Award (2024). ANU Annual Graduate School in Mathematical Aspects of Data Science Participant Support (2024). SEAS Travel Grant, Princeton University (2024, 2025).

Publications

Journal papers

- Roxanne He and Jackie Lok. "On Approximating the Potts Model with Contracting Glauber Dynamics". *Advances in Applied Probability* (2025). To appear, pp. 1–38. DOI: 10.1017/apr.2025.10036. arXiv: 2404.18778.
- Jackie Lok and Elizaveta Rebrova. "A subspace constrained randomized Kaczmarz method for structure or external knowledge exploitation". *Linear Algebra and its Applications* 698 (2024), pp. 220–260. DOI: 10.1016/j.laa.2024.06.010. arXiv: 2309.04889.

Conference papers

- Jackie Lok, Rishi Sonthalia, and Elizaveta Rebrova. "Error dynamics of mini-batch gradient descent with random reshuffling for least squares regression". *Proceedings of the 36th International Conference on Algorithmic Learning Theory (ALT 2025)*. Vol. 272. PMLR, 2025. arXiv: 2406.03696.

Preprints

- Toby Anderson, Max Collins, Jamie Haddock, Jackie Lok, and Elizaveta Rebrova. "Beyond Expectation: Concentration Inequalities for Randomized Iterative Methods" (2025). arXiv: 2511.20877.
- Jackie Lok and Elizaveta Rebrova. "Subspace-constrained randomized coordinate descent for linear systems with good low-rank matrix approximations" (2025). arXiv: 2506.09394.
- Rishi Sonthalia, Jackie Lok, and Elizaveta Rebrova. "On Regularization via Early Stopping for Least Squares Regression" (2024). arXiv: 2406.04425.

Miscellaneous

- Jackie Lok. *Markov chains, mixing times, and cutoff*. Honours thesis. 2020.

Talks and Presentations

- 2026 Joint Mathematics Meetings, Washington D.C.: "Efficient linear solvers for approximately low-rank linear systems" Jan 2026
- 2025 Mid-Atlantic Numerical Analysis Day, Temple University: "Sketch-and-Project Solvers for Linear Systems with Low-Rank Structure" Nov 2025
- Simons Institute Workshop on Linear Systems and Eigenvalue Problems, UC Berkeley: "Subspace-constrained randomized coordinate descent for linear systems with good low-rank matrix approximations" (Poster) Oct 2025
- The Third Joint SIAM/CAIMS Annual Meetings (AN25), Montréal, Canada: "Subspace-constrained Sketch-and-project Solvers for Linear Systems with Low-rank Structure" Jul 2025
- NLA group meeting, University of Oxford, UK: "Subspace-constrained sketch-and-project solvers for linear systems with low-rank structure" Jul 2025
- The 36th International Conference on Algorithmic Learning Theory (ALT 2025), Milan, Italy: "Error dynamics of mini-batch gradient descent with random reshuffling for least squares regression" Feb 2025
- Conference on Mathematical Theory of Deep Neural Networks (DeepMath 2024), Philadelphia: "Error dynamics of mini-batch gradient descent with random reshuffling for least squares" (Poster) Nov 2024
- CUNY Graduate Center Harmonic Analysis & PDE Seminar: "A subspace constrained randomized Kaczmarz method" Nov 2024
- SIAM Conference on Mathematics of Data Science (MDS24), Atlanta: "A Subspace Constrained Randomized Kaczmarz Method for Structure or External Knowledge Exploitation" (Poster) Oct 2024

- NSF CompMath PI Meeting, University of Washington, Seattle: "A Subspace Constrained Randomized Kaczmarz Method for Structure or External Knowledge Exploitation" (Poster) Jul 2024
- Graduate student probability reading group, Princeton University:
  "Concentration for Random Matrix Products" Oct 2023
  "Matrix Concentration Inequalities via the Method of Exchangeable Pairs" Nov 2022
- Honours presentation, UNSW Sydney: "Mixing times of Markov chains and the cutoff phenomenon" Nov 2020

Teaching

Princeton University, *Princeton, NJ, USA*

Teaching Assistant (TA), ORFE Department

Responsible for delivering weekly precepts, holding office hours, grading problem sets and exams, and general course admin.

- ORF 405: Regression and Applied Time Series Fall 2025
- ORF 387: Networks Spring 2025
- ORF 526: Probability Theory Fall 2024
- ORF 387: Networks Spring 2024
- ORF 363: Computing and Optimization Fall 2023
- ORF 350: Analysis of Big Data Spring 2023
- ORF 387: Networks Fall 2022

University of Melbourne, *Melbourne, Australia*

Academic Tutor, School of Mathematics and Statistics

Responsible for delivering weekly tutorials and marking assessments.

- MAST20004: Probability Semester 1 2021

UNSW Sydney, *Sydney, Australia*

Academic Tutor, School of Risk and Actuarial Studies

Responsible for delivering weekly tutorials, marking exams and assessments, developing course materials, holding student consultations, and providing general course support.

- ACTL3162: General Insurance Techniques Term 3 2020
- ACTL2102: Foundations of Actuarial Models Term 2 2020
- ACTL2111: Financial Mathematics for Actuaries Term 1 2020
- ACTL1101: Introduction to Actuarial Studies Term 3 2019
- ACTL2102: Foundations of Actuarial Models Term 2 2019
- ACTL3141: Actuarial Models and Statistics Term 1 2019

Other Activities

Mentoring

- McGraw Graduate Teaching Fellow, *Princeton University* 2024–
- ReMatch mentor, *Princeton University* 2023
- ORFE Senior Thesis Writer's Group co-leader, *Princeton University* 2022–2023
- Drop-in Centre tutor, *School of Mathematics and Statistics, UNSW Sydney* 2020

Reviewing

IMA Journal of Numerical Analysis, Linear Algebra and its Applications, SIAM Journal on Matrix Analysis and Applications

Work Experience

icare, Actuarial Services Intern, *Sydney, Australia* Aug 2018–Feb 2019

Supported the provision of actuarial advice and analysis for the NSW state insurer. Assisted with the reporting and valuation of outstanding claims liabilities, scenario analysis, preparation of financial budgets, claims experience monitoring, and the assessment of data quality and integrity.

Suncorp Group, Natural Perils Pricing Intern, *Sydney, Australia* Feb 2018–Aug 2018

Collaborated in the research and development of a new natural peril pricing model in Python using analytical and machine learning techniques with insurance and geospatial datasets. Developed interactive tool using SAS and Python to identify and visualise exposure concentration risks as part of an automated monitoring pipeline.

MetLife Australia, Capital and Valuation Intern, *Sydney, Australia* Nov 2016–Feb 2017

Assisted with financial reporting, reserving and scenario analysis for group life insurance.

Languages and skills

Languages

English (native), Cantonese (fluent), Mandarin (beginner), German (beginner).

Computing

- Proficient with Python. Experience with other programming languages including Julia, R, Java, MATLAB, SQL, and SAS.
- Competent with LaTeX.
- Experience with Microsoft Excel, Word, and PowerPoint.

Online courses

- Probabilistic Graphical Models Specialization (Coursera, Stanford University) 2021
- Deep Learning Specialization (Coursera, DeepLearning.AI) 2021
- Machine Learning (Coursera, Stanford University) 2018